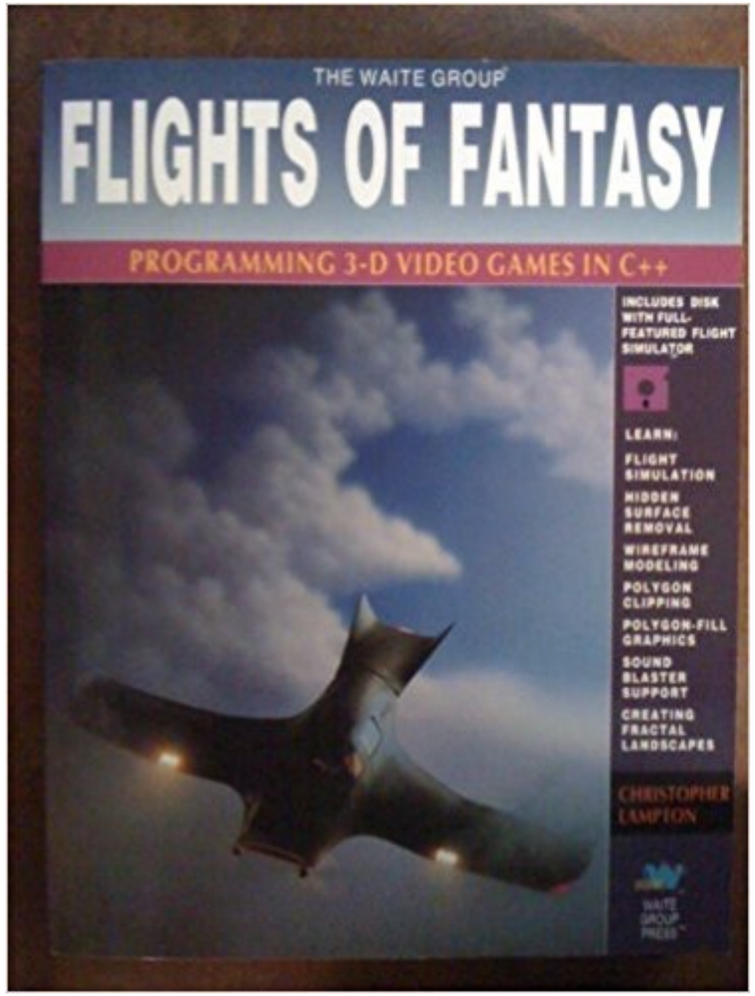
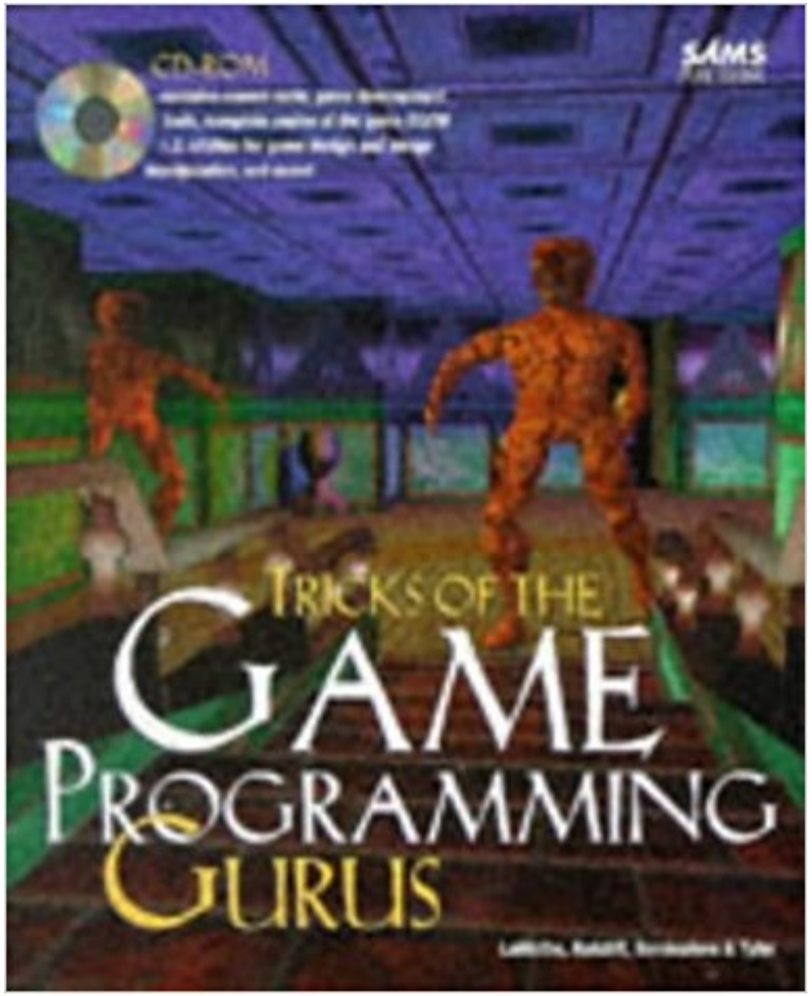
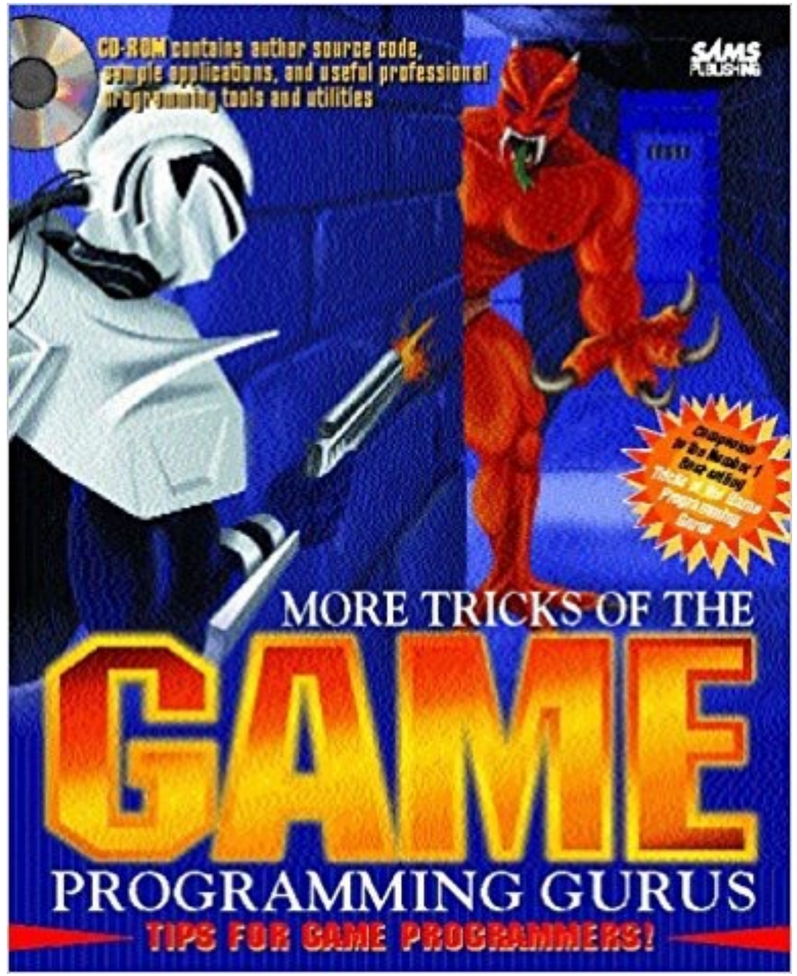
GAME PROGRAMMING
- BASICS -

A. Ricci, DISI - University of Bologna - Italy
a.ricci@unibo.it

SHORT “PERSONAL” PREAMBLE: BACK TO ~1993/1994







NOWADAYS

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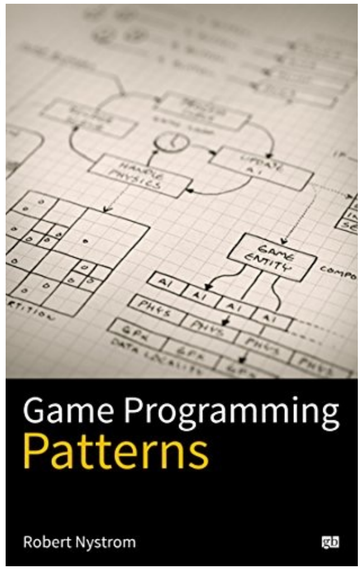
OBJECTIVES

- Introducing some basic game programming aspects, as an application domain where to apply concepts learnt in OOP course (& other bachelor courses)
  - Java as a programming language
  - patterns, software architectures, idioms..
- Method
  - incremental development of a very simple game in Java *("Roll a ball")* in some basic steps, by using some basic game programming patterns

MAIN REFERENCE: GPP

- Reference book
  - **[GPP] Robert Nystrom, “Game Programming Patterns”, 2014**
  - …with some personal adaptations and extensions



GENERAL ASPECTS

- Game architecture: structure and dynamics
- Game world and object modeling
  - structure, behaviour
- Kinematics and dynamics
  - collisions, physics
- Input control & interaction with the physical environment
- Graphics, animation, audio
  - rendering the scene, effects
- AI
- Networking & multi-player

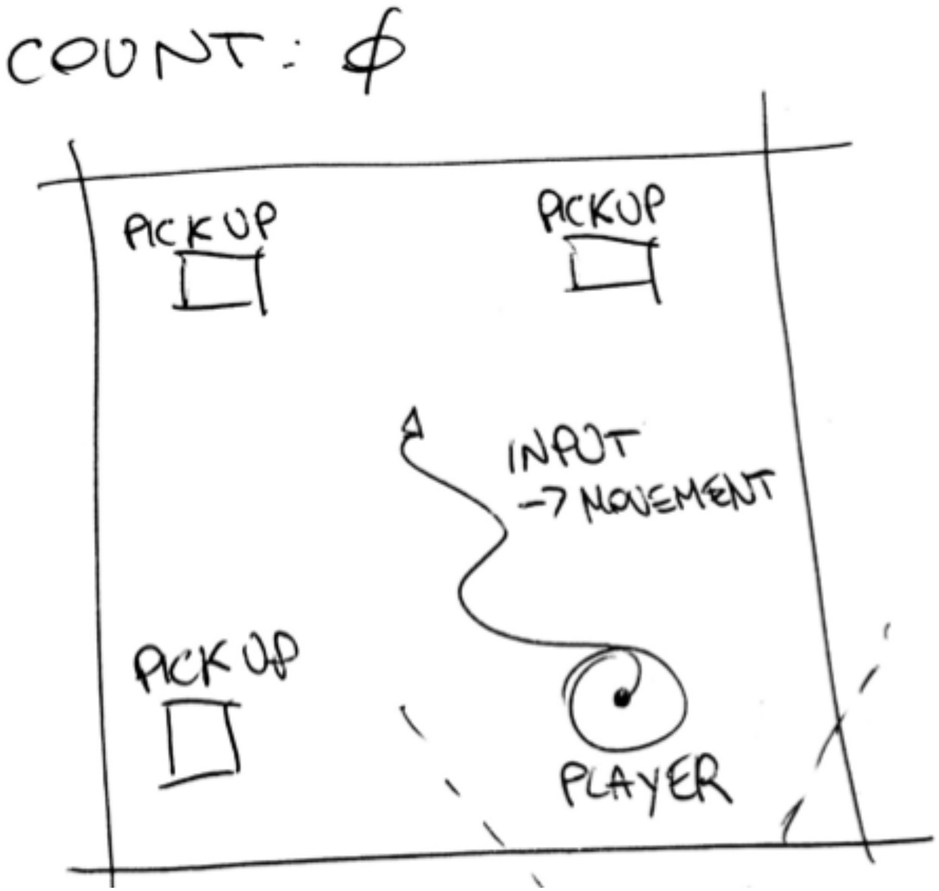
GAME ENGINES

- ***Frameworks*** factorizing functionalities/services
  - ~OS for developing and running games, isolating the game from the HW on which it is running
  - *middleware* perspective
    - vs. libraries
- Many examples
  - http://en.wikipedia.org/wiki/List_of_game_engines

PROGRAMMING A GAME IN FEW STEPS

“ROLL A BALL”

- A *ball* rolling through a 2D environment, moved by the player
- Objective:
  - collect the *pick-up objects* as fast as possible, avoiding borders



(Pierluigi Montagna’s sketch)

PATTERNS IN THIS SHORT JOURNEY

- GAME LOOP
- COMMAND
- OBSERVER
- EVENT QUEUE
- COMPONENT

STEP #0:
MEET THE **GAME LOOP**

THE GAME LOOP

- Control loop ruling the execution of a game
  - *"Decouple the progression of game time from user input and processor speed"* [GPP, CH 9]
- Evolution ~ sequence of frames

```
while (true) {
  processInput()
  update()
  render()
}
```

SYNCH WITH REAL TIME

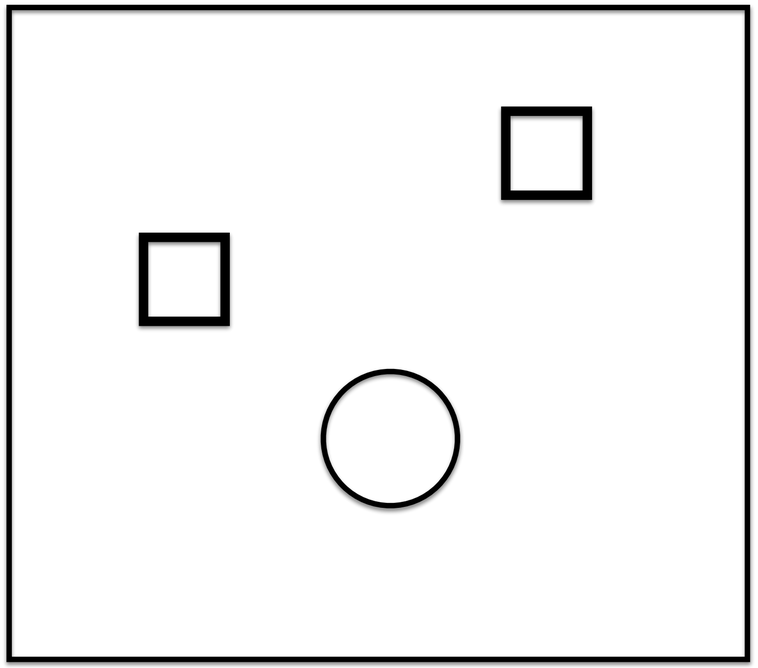
- Choosing a time step to advance the game depending on how much real time elapsed since the previous frame
  - the game runs at the same rate in spite of the specific HW…
  - players with more powerful HW gains in gameplay smoothness
- Check the code: Step #0

```
double lastTime = getCurrentTime()
while (true) {
  double current = getCurrentTime()
  double elapsed = current - lastTime
  processInput()
  update(elapsed)
  render()
  lastTime = current
}
```

STEP #1:
START MODELING THE GAME

START MODELING THE GAME

- OOP modeling of Roll-a-Ball game
  - **GameObject**
    - Ball, PickUpObj
  - **World**
- Basic *separation of concerns*
  - model/graphics
- Check the code: Step #1

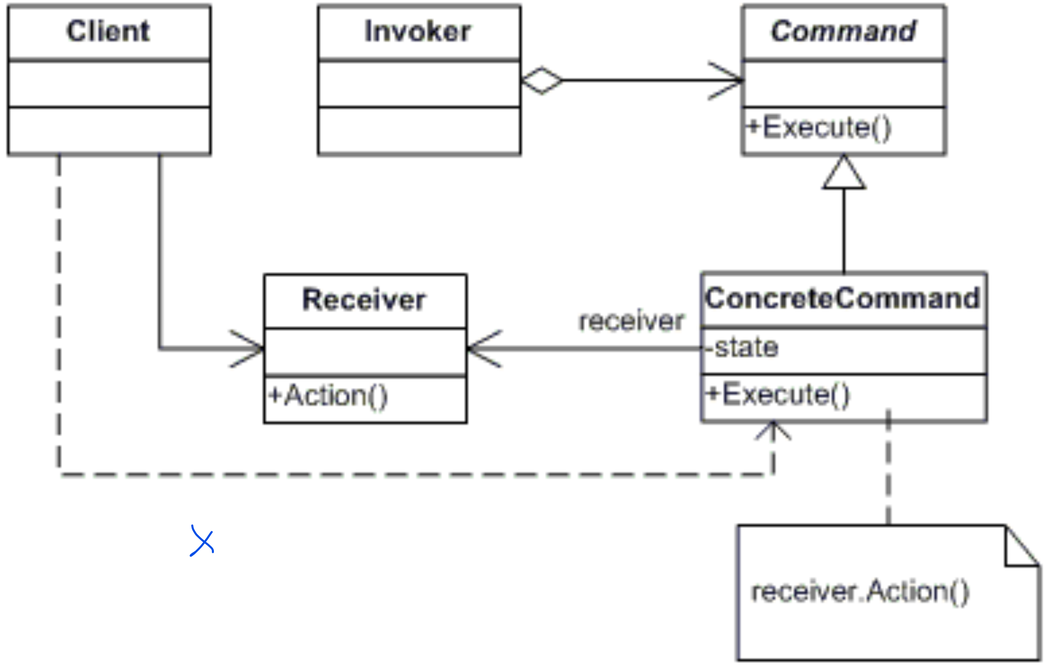


STEP #2:

MEET THE **COMMAND** PATTERN

INPUT PROCESSING

- **Command** pattern
  - *"Encapsulate a request as an object, thereby letting users parametrize clients with different requests, queue or log requests…"* [GPP, CH2]
- Async processing - event listening
  - async keyboard events => commands
- Check the code: Step #2



STEP #3
MEET COLLISIONS

A LITTLE BIT OF PHYSICS: COLLISIONS

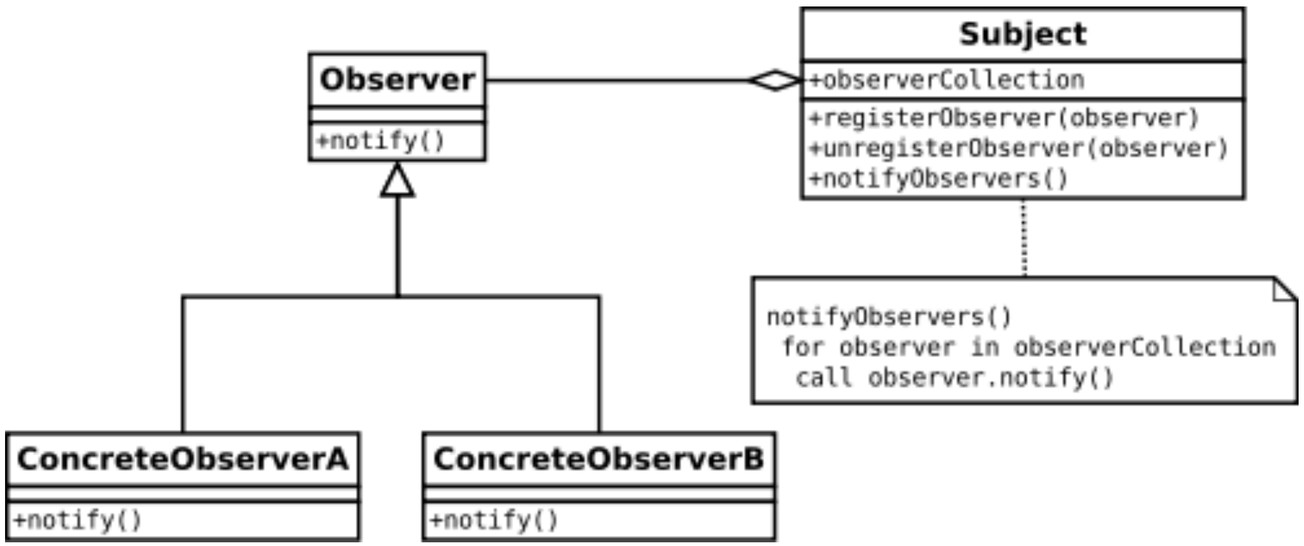
- Simple collision management
  - Ball + PickUpObj, Ball + borders
  - meet Bounding Boxes
- Check the code: Step #3

STEP #4

GAME STATE + EVENTS

GAME STATE + EVENTS

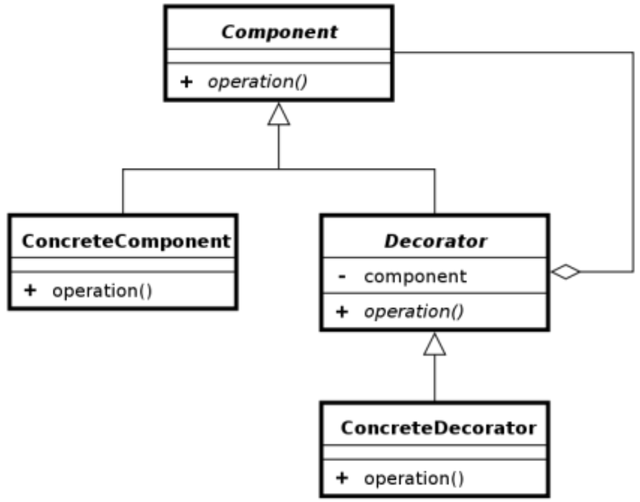
- Introducing the game state, that includes a score
  - score +1 for PickUp, -1 for borders
- Decoupling collision management from the effects
  - **Observer** pattern
    - async variant
- Check the code: Step #4



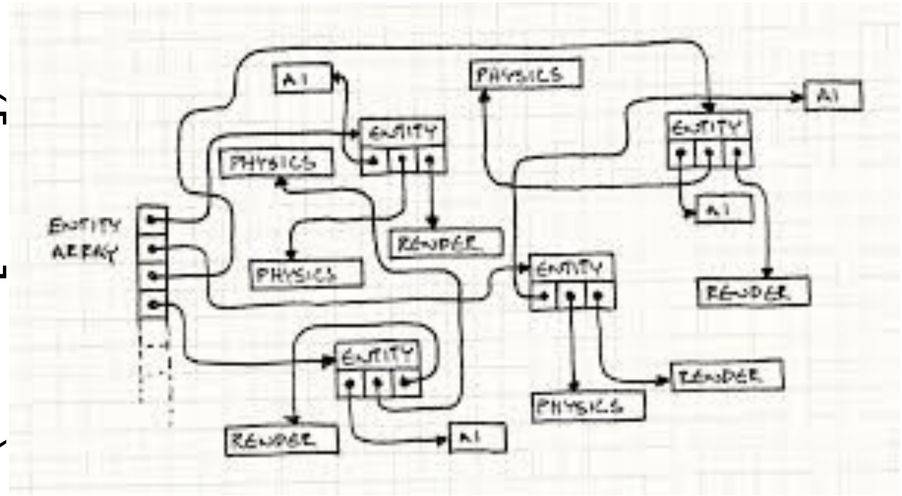
STEP #5
MEET THE **COMPONENT** PATTERN

COMPONENT PATTERN

- **Component** pattern
  - *"Allow a single entity to span multiple domains without coupling the domains each other"* [GPP, CH 14]
- Major refactoring
  - full separation of concerns based on domains (physics, graphics, input)
  - pure Game Objects + Game Object Factory
- Check the code: Step #5



(from [GPP])



STEP #6
GAME OVER

GAME OVER

- Back to the game objective: conditions to end the loop
  - based on GameState
- Game state based rendering
- Check the code: Step #6

GAME OVER

FURTHER STEPS

- Step 7: Mosquito Player
  - first dumb AI player *(MosquitoAIInputComponent)* controlling the ball, random movement
  - development: **State** pattern & AI strategies
- Step 8: Refining physics with accelerations
  - keyboard-based player*(PlayerInputComponentWithAcc)* that produces movements with accelerations
- … *(time exhausted)*

FURTHER STEPS
(NOT DISCUSSED, BUT CODE AVAILABLE)

- Step 9: Multiple Balls
- Step 10: Multi-player
- Step 11: Multi-player with remote users
- Step 12: Multi-player with remote users with view